

Experiment No: 4

Title:

To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to set up a build Job.

Aim:

To study the concept of Continuous Integration (CI), install Jenkins, configure the environment, and create a basic build job using Jenkins with build tools such as Maven, Ant, or Gradle.

Theory:

Continuous Integration (CI) is a software development practice in which developers frequently integrate code changes into a shared repository. Each integration triggers automated build and testing processes to detect errors early in the development cycle. This ensures improved software quality and faster delivery of applications.

Jenkins is an open-source automation server widely used for implementing Continuous Integration. It helps automate building, testing, and deployment of applications. Jenkins integrates easily with version control systems such as Git and supports multiple build tools like Maven, Ant, and Gradle.

Key components of Continuous Integration include Version Control Systems, automated builds, automated testing, and CI servers such as Jenkins. By configuring Jenkins with build tools, developers can automate compilation and execution tasks efficiently.

Requirements:

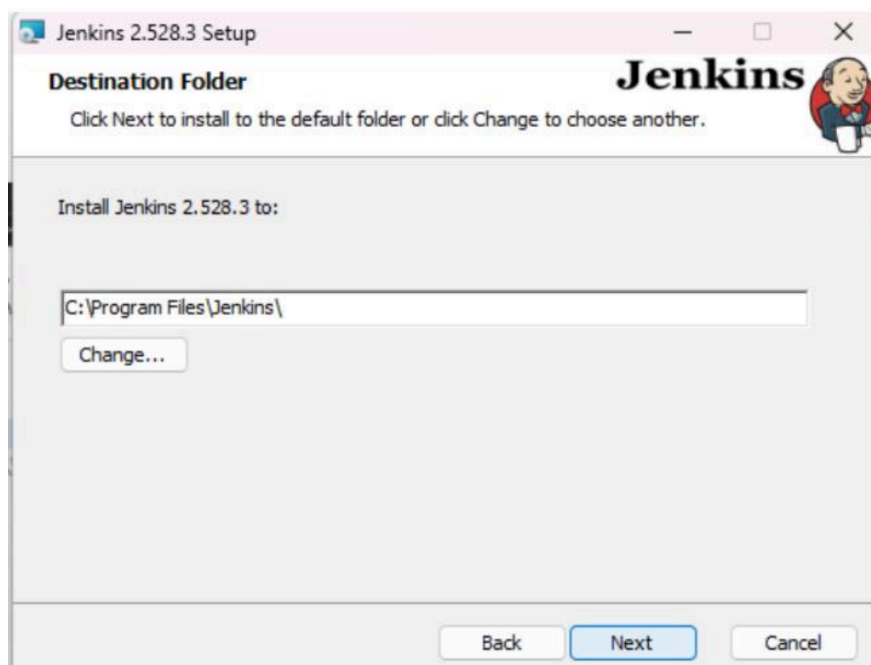
- Jenkins installer
 - Java Development Kit (JDK)
 - Web browser
 - Internet connection
-

Procedure:

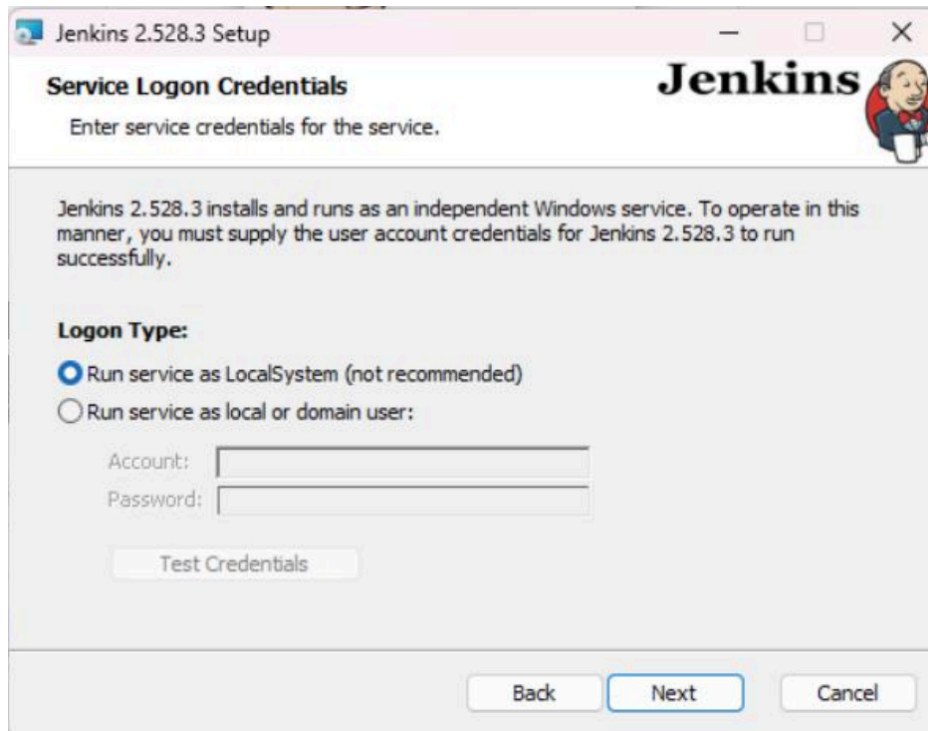
Step 1: Download and launch Jenkins installer from the official Jenkins website.



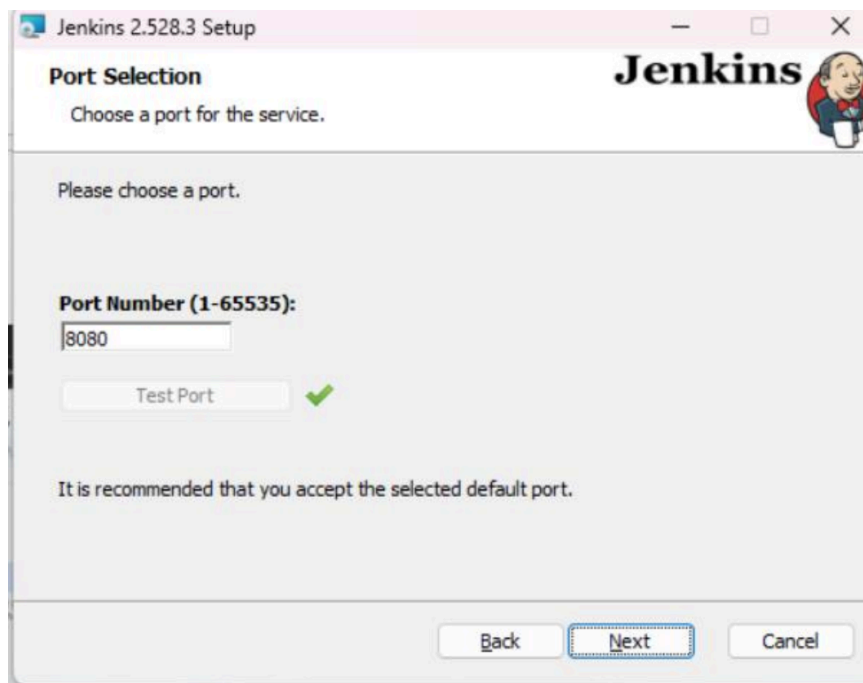
Step 2: Select installation destination folder and proceed with setup.



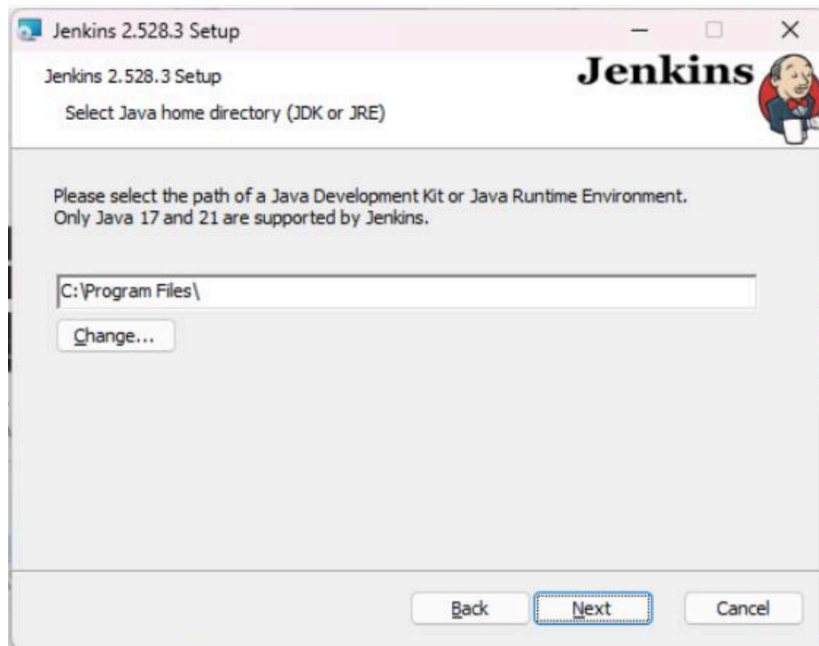
Step 3: Configure service login credentials for Jenkins service.



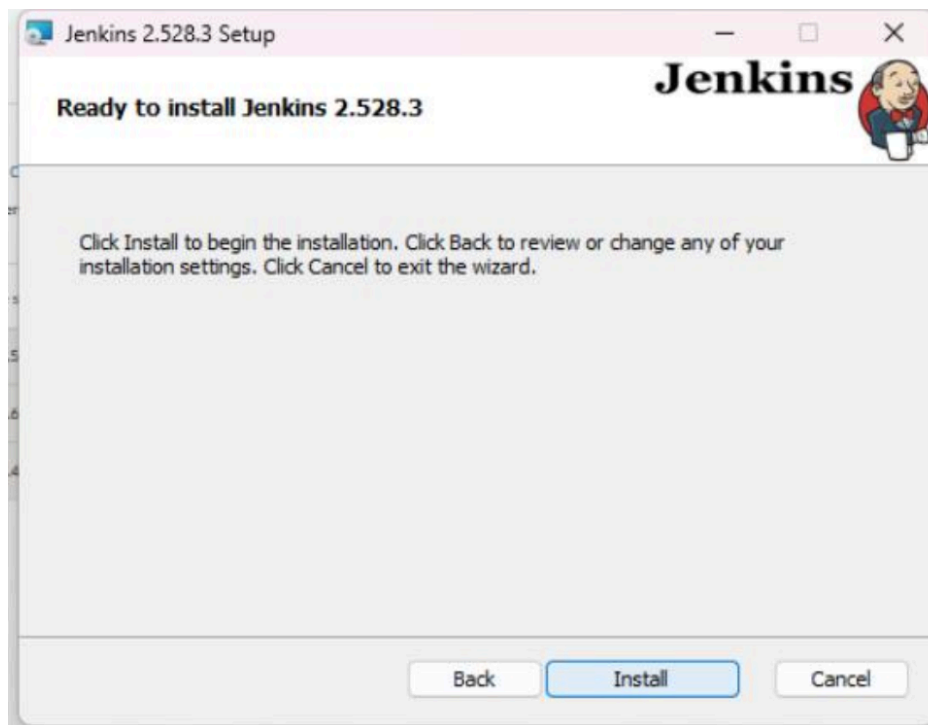
Step 4: Configure default port number (8080) for Jenkins server.



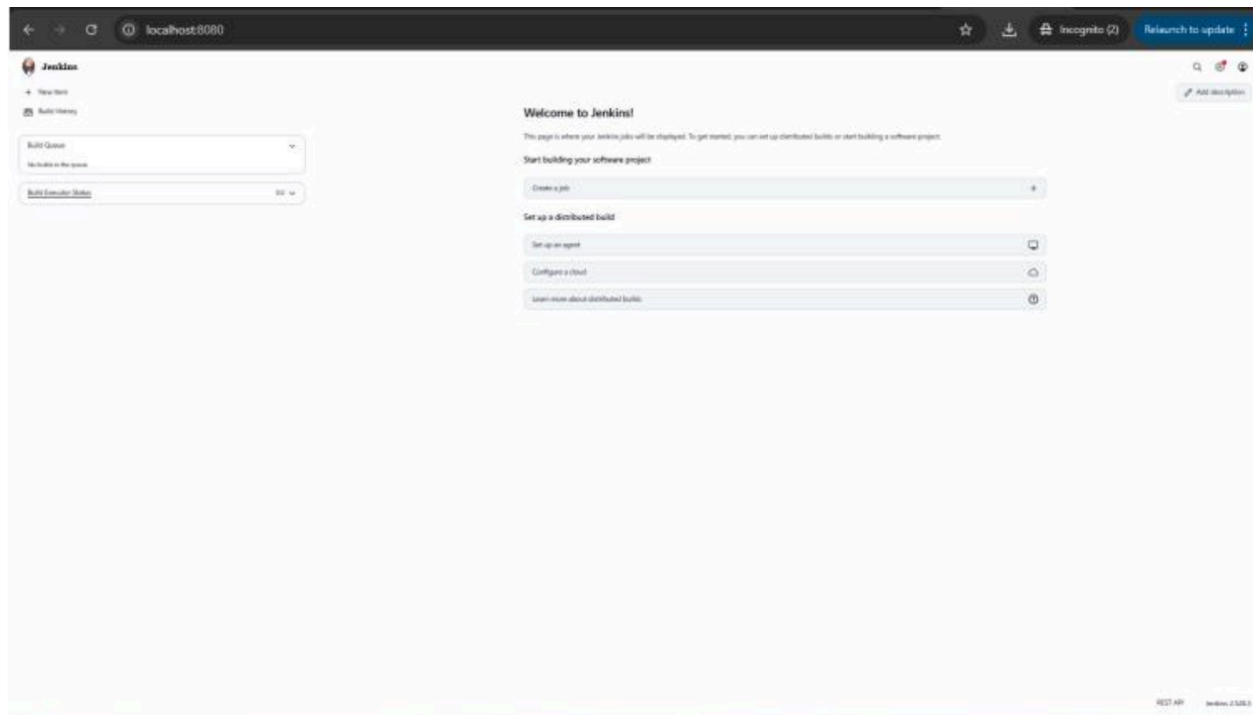
Step 5: Select Java Development Kit (JDK) installation path required for Jenkins execution.



Step 6: Confirm installation settings and begin Jenkins installation process.



Step 7: Open Jenkins dashboard in browser using localhost:8080 after installation completion.



Step 8: Create a new build job in Jenkins by selecting “New Item” and choosing project type such as Freestyle project or Pipeline project.

Step 9: Configure build tool settings such as Maven/Ant/Gradle and execute the build job.

Result:

Successfully installed and configured Jenkins Continuous Integration server and created a build job using Jenkins with build tool support.

Conclusion:

Thus, Continuous Integration was successfully implemented by installing and configuring Jenkins and setting up a build job using Maven/Ant/Gradle tools.